Leakage, Label Noise, and Overestimated Performance in T-Cell Epitope Immunogenicity Prediction

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Abstract

Computational predictors of CD8⁺ T-cell epitope immunogenicity report ROC-AUC values of 0.85–0.90 on data derived from the Immune Epitope Database (IEDB), and are used to prioritise candidates for neoantigen vaccines. We ask whether the labels these models are trained and evaluated on can support such values. From 65,732 human MHC-I T-cell assay records we identify 2,914 peptide–HLA pairs assayed independently by two or more publications, and measure how well one laboratory's outcome predicts another's on the same pair. Agreement is 72.2%, Cohen's κ = 0.273, and the equivalent AUC is 0.637 (95% CI 0.628–0.648); restricting to the least-selected subset, pairs assayed by exactly two studies, gives 0.681. A repeat experiment observes the biology directly and is therefore the most informative predictor obtainable, so reported performance exceeds what these labels can support. We then identify three specific benchmark-construction choices that produce the excess. First, splitting data by peptide — the field's standard precaution — performs indistinguishably from not splitting at all (IEDB: 0.826 vs 0.840 random; an independent dataset: 0.862 vs 0.857), whereas grouping by source publication reduces AUC to 0.607. This follows from study-level label structure: one all-negative study supplies 37% of all pairs, 75% of studies report a single outcome class, and a baseline using allele identity alone, with no peptide information, reaches 0.752 — above a widely used published predictor at 0.647. Second, two published predictors were trained on 37.5% and 26.5% of our evaluation pairs, and score 0.07 and 0.26 AUC lower on pairs they had not seen. Third, defining negatives as untested candidates rather than assayed negatives raises AUC from ≈0.55 to ≈0.97 for the same nominal task. Under leakage-free evaluation, measuring within study and within allele across 205 groups, the best model we construct reaches 0.573 (95% CI 0.544–0.603) — 84% of the conservative reliability bound — and the original convolutional architecture reaches 0.544, a difference that is not significant. We release the benchmark, both split schemes, all model predictions, and a protocol for reporting.

Keywords: T-cell epitope · immunogenicity prediction · data leakage · benchmark design · label reliability · reproducibility · IEDB

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1. Overview

Determining which peptide–MHC complexes elicit a CD8⁺ T-cell response is central to neoantigen vaccine design and adoptive cell therapy. Because experimental screening is slow and costly, computational predictors trained on IEDB have proliferated, and recent tools report ROC-AUC values between 0.85 and 0.90.

Those figures sit uneasily beside two established facts.

The target is not a function of the inputs. Immunogenicity does not depend on peptide and allele alone. It depends on the individual's T-cell receptor repertoire, prior antigen exposure, the remainder of their HLA genotype, and the tissue context. Two individuals carrying the same restricting allele can respond differently to the same peptide. Any model given only a peptide and an allele therefore has a ceiling below 1.0, whatever its architecture.

The measurements are variable. International proficiency panels run by the Cancer Vaccine Consortium and the CIMT Immunoguiding Programme found inter-laboratory variation of roughly 35-fold for ELISpot, 20-fold for intracellular cytokine staining and 100-fold for tetramer assays, and identified laboratories unable to detect all predefined responses "even on a yes/no basis" [1].

The immunology community responded with harmonisation guidelines, which reduce variation in prospective, coordinated studies [1]. They cannot retroactively harmonise a database aggregating decades of heterogeneous protocols. The consequence of that heterogeneity for machine-learning benchmarks built on IEDB has not, to our knowledge, been measured. Predictors instead handle label discordance by convention: PredIG, for example, discards peptide–HLA pairs whose outcomes disagree across experiments [8].

Prior work. Related problems have been identified. Lee et al. documented HLA bias in immunogenicity predictors and showed that cross-species splits degrade performance [3]. Benchmarks of TCR–epitope predictors report near-chance performance on unseen epitopes, and warn that external negatives "introduce batch-like confounders, causing models to learn dataset-specific artifacts" [9]. Kapoor and Narayanan surveyed 17 scientific fields and found leakage affecting 294 papers, showing in a case study that complex models claimed to beat logistic regression did not do so once leakage was corrected [6].

What has been missing is a quantitative account of what the labels themselves can support, and of which specific construction choices produce the gap between that bound and reported performance. This paper supplies both. Our contributions are:

1. **A measured reliability bound** for IEDB immunogenicity labels, with sensitivity

analyses over aggregation rule, retesting frequency, assay family and HLA locus.

2. **A demonstration that peptide-level splitting is ineffective**, replicated on an

independent dataset, and identification of source publication as the grouping unit that matters.

3. **A contamination audit** of two widely used published predictors against our evaluation

set.

4. **A leakage-free performance estimate**, an architecture comparison showing the choice is

immaterial, and a worked example of a feature that standard validation endorses and grouped validation rejects.

2. Scope and definitions

Scope. We examine benchmarks for CD8⁺ T-cell epitope immunogenicity built from IEDB assay records, where a model receives a peptide and an MHC-I allele and predicts whether a T-cell response was observed. We do not examine MHC binding or ligand presentation prediction, which are better-posed problems with more reproducible labels and where reported performance is correspondingly higher [9]. Nor do we examine patient-specific neoantigen prioritisation pipelines, which use tumour sequencing, expression and clonality in addition to sequence [4]; §3.7 shows why conflating the two inflates apparent performance.

Grouping unit. A split is defined by the unit held out. Splitting *by peptide* means no peptide appears in both training and test data; splitting *by study* means no record from a given publication appears in both. Leakage here is not duplicated rows but shared structure: records from one publication share a cohort, a protocol and often an outcome, so a model can identify the source rather than the biology. We use "grouping unit" throughout to name the choice, because it is the choice that determines the reported number (§3.4).

Label reliability. We define the reliability of a label set as the agreement between two independent measurements of the same quantity — here, two publications assaying the same peptide–HLA pair. Expressed as an AUC, this is the score a repeat experiment achieves when used to predict another experiment's outcome. Because a repeat experiment observes the biology rather than inferring it from sequence, it is the most informative predictor obtainable for that pair.

What reliability does and does not bound. Reliability bounds what can be *measured* against a given label set; it is not a bound on the attainable accuracy of immunogenicity prediction in principle. When annotators carry independent noise, a well-specified model can exceed their mutual agreement [7]. We return to this in §4.

Operational task. We take the task that matters for vaccine design to be ranking candidates *within one experiment*: given the peptides a laboratory is considering for one patient or one antigen, which should be screened first. This motivates reporting within-study and within-allele AUC alongside pooled figures (§7.5).

3. Results

3.1 One study supplies a third of the data, and most studies report a single outcome

After retrieval and cleaning (§7.1), the benchmark comprises 31,274 unique peptide–HLA pairs, 31.7% positive, drawn from 1,365 publications. Its composition is not uniform (**Figure 1**). A single study contributes 11,486 pairs — 37% of the dataset — and every one of its outcomes is negative. Across all studies, 75% report only one outcome class.

Study identity is therefore highly informative about the label before any peptide is examined. This observation motivates the analyses that follow.

3.2 Immunogenicity labels reproduce poorly between laboratories

Restricting to the 2,914 pairs assayed by two or more distinct publications, and taking each study's majority outcome, we asked how well one study's result predicts another's on the same pair (§7.3).

Quantity	Value (95% CI
Agreement	0.722 (0.714–0.
Cohen's κ	**0.273** (0.255–0.
AUC	**0.637** (0.628–0.

A κ of 0.273 denotes fair agreement only. Because a repeat experiment observes the underlying biology rather than inferring it from sequence, it is the most informative predictor obtainable for a given pair; its agreement with another experiment bounds what can be *measured* against these labels.

3.3 The reliability estimate is robust to how it is measured

Three threats to this estimate were tested (**Figure 3**).

Aggregation rule. Majority vote 0.637; any-positive 0.652; ties-to-positive 0.646; ties-to-negative 0.556; and raw records with no aggregation at all 0.594. The range is 0.556–0.652, and the least-processed variant falls below the headline figure.

Selection bias. Pairs are not retested at random. Reproducibility declines with retesting frequency: 0.681 for pairs assayed by exactly two studies (n = 1,806), 0.531 for exactly three, 0.542 for four or more. We therefore adopt **0.681 as a conservative bound**. Separately, repeat-tested pairs are 72.5% positive against 26.5% for singly-tested pairs: the decision to retest is driven by the previous result.

Slice dependence. By assay family, ELISpot 0.703, ICS 0.653, multimer/tetramer 0.604; by locus, HLA-A *0.594, HLA-B* 0.775. Every slice falls far below reported performance. The ordering across assay families independently reproduces the proficiency-panel finding that tetramer assays vary more than ELISpot [1].

3.4 Peptide-level splitting does not prevent leakage

Holding out data by peptide is the standard precaution in this field. We compared grouping units using one model class throughout, so that only the split changes (**Figure 2**).

Grouping unit	IEDB (this study)
Random	0.840
Peptide	**0.826**
Antigen	—
Source (study / organism)	**0.607**

On the independent dataset, peptide-grouping scores *above* random splitting. In both datasets, widening the grouping unit to the data source collapses performance. The precaution the field takes does not address the structure that inflates its results.

3.5 A baseline that cannot see the peptide outscores published predictors

On the held-out test split, an allele-prior baseline — predicting each allele's training-set positive rate, using no peptide information whatsoever — reaches 0.752. BigMHC_IM reaches 0.647 and MixMHCpred 0.559 on the same pairs (**Figure 4**).

A peptide-blind baseline exceeding published predictors is diagnostic: on this benchmark, allele identity and dataset structure carry more signal than the peptide sequence does.

3.6 Published predictors were trained on the evaluation data

Intersecting our evaluation pairs with released training sets (§7.5) shows substantial overlap, and performance falls sharply outside it (**Figure 5**).

Predictor	Overlap	Seen	Unseen
DeepImmuno [2]	37.5% of test peptides	0.835	0.763
BigMHC_IM [9]	26.5% of lung pairs (exact pair)	0.807	**0.549**

On 778 lung-cancer-associated pairs unseen by both, every published predictor's interval includes chance: DeepImmuno 0.528 (0.475–0.581), BigMHC 0.502 (0.438–0.563), MixMHCpred 0.446 (0.384–0.503), allele prior 0.464 (0.414–0.513).

3.7 The definition of a negative changes the task by ≈0.4 AUC

Using the harmonised multi-cancer resource of Müller et al. [4]: across 99 patients there are 1,780,481 candidate neo-peptides, of which 421,017 (23.6%) were experimentally screened and 178 were immunogenic.

Ranking immunogenic peptides against *all* candidates — 76% never tested — gives per-patient AUC ≈ 0.97. Discriminating assayed positives from assayed negatives gives ≈ 0.55. Both are published as immunogenicity prediction. Most "negatives" in the former are peptides nobody screened, overwhelmingly non-binders that any filter rejects.

3.8 Leakage-free performance approaches the measured ceiling, and architecture is immaterial

Scoring every model from study-grouped out-of-fold predictions, within study and allele, across 205 groups, bootstrapping over studies (**Figure 6**):

Training scheme	AUC	95% CI
Ranking objective, no engineered features	**0.573**	0.544–0.603
Ranking objective, with features	0.557	0.526–0.588
CNN (original Conv2D architecture)	**0.544**	0.516–0.572
Study reweighting	0.548	0.511–0.584
Pointwise, no features	0.538	0.511–0.566
Pointwise baseline	0.532	0.504–0.560

All intervals clear chance. The best result is 84% of the conservative reliability bound (0.681). The convolutional architecture, given identical pretraining, folds and seeds, differs from the attention model by +0.013 (95% CI −0.017 to +0.043, not significant): **the architecture choice does not matter here.** The only comparison that reaches significance is the ranking objective over pointwise training (+0.035, CI +0.001 to +0.069), and its lower bound is close enough to zero that we report it as marginal.

Simple study-aware reweighting recovers much of the loss on pooled metrics (0.623 → 0.700), and is a cheaper mitigation than a bespoke objective.

3.9 A worked example: a feature that standard validation endorses and grouped validation rejects

Self-similarity to the human proteome appears strongly predictive. Pooled across lung-cancer-associated pairs it reaches AUC 0.671 (CI 0.630–0.712), and under peptide-grouped validation it improved within-allele AUC by +0.034.

Under study-grouped validation it provides nothing. Within study, its AUC is 0.483, with **0% of studies above chance**. Withholding the entire feature block changes within-study AUC by +0.005 (CI −0.015 to +0.024) and within-study-and-allele AUC by −0.007, neither significant. The feature separates studies, not peptides.

Of nine feature families tested within study, only two clear chance: mean hydrophobicity 0.550 (0.518–0.582) and aromatic content 0.542 (0.517–0.568), both consistent with known determinants of TCR recognition [8].

4. Discussion

Reported AUCs of 0.85–0.90 exceed both the measured reliability of the labels (0.637; conservatively 0.681) and the performance of any model we could construct under leakage-free evaluation (0.573). The excess is not evidence of superior models. It is produced by three separable choices: splitting by peptide, which is equivalent to not splitting; training-set overlap, present in both audited predictors; and defining negatives as untested candidates, which changes the task.

A diagnostic follows. Because measured AUC is computed against noisy labels, a model recovering only transferable biological signal should score near the label reliability. Systematically exceeding it implies the model is also predicting the noise structure — and we show that structure is study-level.

On the interpretation of agreement as a bound. Inter-annotator agreement is not in general an upper bound on achievable model performance: when annotators carry independent noise, a well-specified model can exceed their mutual agreement [7]. Our claim is therefore about *measured* performance against these labels, not about the attainable accuracy of immunogenicity prediction in principle. This distinction does not weaken the conclusion. If the label noise were random, high measured AUC would be unattainable; that it *is* attainable indicates the noise is systematic and learnable, which is precisely what §3.4–2.5 demonstrate.

Relation to prior work. This is an instance, in computational immunology, of the pattern Kapoor and Narayanan documented across 17 fields [6]. The parallel is close: in their case study, complex models claimed to beat logistic regression and ceased to do so once leakage was corrected; in ours, published predictors are outscored by a baseline that cannot see the peptide. Our setting permits one step theirs did not — because IEDB records which publication produced each measurement, and some pairs were measured more than once, the reliability of the labels can be measured directly rather than assumed. We extend Lee et al. [3], who identified HLA bias and cross-species degradation, by identifying the source publication as the operative grouping unit and by quantifying the label ceiling.

What bounds performance here. At 0.573 against a conservative bound of 0.681, we are at 84% of what the labels permit, leaving ≈0.11 AUC of headroom. The evidence suggests this is not reachable by architecture: the convolutional and attention models are statistically indistinguishable, and methods that do substantially better use information absent from sequence-only benchmarks — tumour expression, clonality, or the patient's own tumour-infiltrating repertoire [4]. Approaches adding TCR sequence without patient context report near-chance performance on unseen epitopes [9]. The constraint is consistent across sequence-based, structure-based and TCR-aware methods.

The implications are therefore about data rather than models. Raising the ceiling requires harmonised assays of the kind proficiency panels have shown to reduce variance [1]; broadening coverage requires more distinct epitopes, currently limited by

peptide-library production; and patient-specific prediction requires patient-specific measurement.

4.1 Recommendations

1. Split by **source publication**, not only by peptide, and state the grouping unit used.
2. Report **within-study** and **within-allele** AUC alongside any pooled figure.
3. Include a **peptide-blind allele-prior baseline**. If it is competitive, the benchmark is

measuring dataset structure.

4. State the **negative-set definition** explicitly; untested candidates and assayed

negatives are different tasks.

5. **Audit training-set overlap** before reporting comparative performance.
6. Bootstrap over **studies**, not rows; peptides within a study are not independent.

5. Limitations

- Label reliability is measured on IEDB alone. The independent dataset available to us

lacks publication identifiers, so §3.4 but not §3.2 could be replicated externally.

- Repeat-tested pairs are a selected subset. We address this by stratification and adopt the

most favourable bound, but the estimate remains conditional on that subset.

- MixMHCpred's training set is not public and could not be audited for contamination.
- We show our protocol is stricter, not that it predicts clinical success better.

Establishing the latter would require prospective experimental validation.

- Analysis is restricted to 9–10mers and HLA-A/B/C, inherited from the encoding of the

predictor we reimplemented.

- We propose no improved predictor. Our best model is a leakage-free reference point, not a

contribution; its apparent advantage over published tools on the peptide-grouped split (0.832 vs 0.792) is an artifact of the split we argue against.

6. Conclusion

The labels underlying T-cell epitope immunogenicity benchmarks reproduce across laboratories at AUC 0.637 (κ = 0.273; conservatively 0.681). Published predictors report 0.85–0.90. We trace the difference to peptide-level splitting that provides no protection, training-set contamination of 26–38%, and negative-set definitions that alter the task being measured. A leakage-free model reaches 0.573, attaining 84% of what the labels permit, and the original convolutional architecture is statistically indistinguishable from it. Progress in this area is presently limited by measurement reproducibility and epitope coverage rather than by model architecture.

7. Methods

7.1 Data retrieval and cleaning

T-cell assay records were retrieved from the IEDB Query API restricted to MHC class I, human host, and linear peptides of length 9–10 (92,721 records). Records were required to carry a four-digit HLA-A/B/C allele present in our pseudo-sequence table and a canonical twenty-letter peptide, leaving 65,732 records over 32,167 unique pairs. Any *Positive** qualitative measure was mapped to 1 and *Negative* to 0. Records were collapsed to one label per pair by majority vote; 893 exact ties were dropped, giving **31,274 pairs, 31.7% positive**. A further 400,000 MHC binding and ligand-elution records were retrieved under the same constraints for encoder pretraining.

7.2 Split schemes

Two schemes are stored in the same table: *split*/*fold* grouped by peptide, and *study_split*/*study_fold* grouped by *reference_id*. Each uses a 15% grouped hold-out and 10 grouped CV folds.

7.3 Reliability measurement

Pairs assayed in ≥2 distinct publications were identified. Each study contributes its majority outcome for that pair; within-study ties are dropped. Two studies are drawn at random per pair, and we compute agreement, Cohen's κ, and the AUC of study A's label predicting study B's. This is repeated over 200 random pairings; we report the mean and the 2.5/97.5 percentiles. Sensitivity was assessed over four aggregation rules plus an unaggregated variant, over assay families, over HLA locus, and stratified by the number of studies testing a pair.

7.4 Models

Published predictors. DeepImmuno [2] (two-branch Conv2D); BigMHC_IM [9] (seven-network ensemble, presentation-pretrained then transfer-learned); MixMHCpred 3.0 [10] (binding only, included as a control). PRIME 2.0 was obtained but its bundled Windows binary did not execute; it is excluded and documented.

Baselines. An allele prior predicting each allele's training-set positive rate, using no peptide information; gradient boosting on the AAindex/PCA encoding.

This study. A bidirectional cross-attention network (65,987 parameters): learned residue and position embeddings, self-attention within the 10-position peptide and the 46-residue HLA pseudo-sequence, then cross-attention in both directions, a dense trunk and a sigmoid output. Pretrained on the binding corpus (held-out binding AUC 0.958) and fine-tuned on immunogenicity.

Training schemes. Pointwise cross-entropy; dropping the dominant study; study-size reweighting; training on mixed-outcome studies only; and a study-stratified pairwise ranking objective over (immunogenic, non-immunogenic) pairs sharing a study and an allele. The CNN was trained from fresh initialisation — never from the published checkpoint — with identical pretraining, folds, class weighting and seeds.

7.5 Evaluation

Pooled AUC is reported with two stricter quantities: within-allele AUC, computed inside each allele; and within-study and within-study-and-allele AUC, computed inside each source publication. Groups require ≥10 pairs and both outcome classes. Confidence intervals for within-study quantities bootstrap over studies rather than rows. Models are scored from study-grouped

out-of-fold predictions. Published training sets were intersected with our evaluation pairs by exact peptide and by exact peptide–HLA pair.

Figures

Figure 1 — Composition of the benchmark. (a) Pairs per study, ranked, log scale; the largest study contributes 11,486 pairs (37% of the dataset) and is entirely negative. (b) Studies by outcome composition: 75% report a single outcome class.

Figure 2 — The grouping unit determines the reported number. Pooled AUC under progressively wider grouping units, for (a) IEDB and (b) the independent TRAP dataset. Random and peptide-grouped splits are indistinguishable in both.

Figure 3 — Measured reliability of the labels, and its robustness. AUC of one study's outcome predicting another's on the same pair, across aggregation rules, retesting strata, assay families and HLA loci. Shaded band marks published claims (0.85–0.90).

Figure 4 — Predictors on the held-out test split. Pooled and within-allele AUC. The allele-prior baseline uses no peptide information.

Figure 5 — Training-set overlap. AUC of two published predictors on pairs they were trained on versus pairs they had not seen.

Figure 6 — Leakage-free performance against the measured ceiling. Within-study, within-allele AUC with intervals bootstrapped over studies, against the measured label reliability band and the range of published claims.

Data and code availability

All analyses are reproducible from *docs/PIPELINE.md*, and *notebooks/Reproduce_Manuscript.ipynb* reproduces every number and figure. Code is organised as *src/data* (retrieval, cleaning, splits), *src/models* (architectures, training) and *src/analysis* (all reported analyses). The curated benchmark, both split schemes and all model predictions are released with the repository.

External resources: IEDB Query API; UniProt UP000005640; TRAP [3]; the harmonised datasets of Müller et al. [4]; the public releases of DeepImmuno [2], BigMHC [9] and MixMHCpred [10].

Author contributions

Both authors contributed to study design and manuscript preparation. Correspondence and submission are handled by the corresponding author, to be designated prior to submission.

Competing interests

The authors declare no competing interests.

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